
CS246: Database Management Systems Lab

Week # 12 (1 Questions, 100 Marks)

Lab session: AL1

Held on: Monday, 06-Apr-2026

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission deadline: 17:00 hrs, 06-Apr-2026

Submissions accepted from: 16:45 hrs, 06-Apr-2026

Submissions accepted till: 18:00 hrs, 06-Apr-2026

Instructors Dr. V. Vijaya Saradhi

Every one minute delay after 17:30 hrs will result in one mark deduction

Please do not quote Network issues or technical issues after 17:30hrs

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Question 1: (100 points)

Using C++ and the MySQL C API interface functions, write a program to evaluate the responses of 1000 JEE ADV candidates against the provided solutions and generate subject-wise marks and total marks. You are given:

1. Answer key file: `key.csv`. Each row contains: `question_number`, `subject`, `question_type`, `correct_answer`, `maximum_marks`
2. Responses file containing 1000 candidates responses: `jee_candidates_1000.csv`
Each row contains one registration number followed by:
 - (a) 16 answers of MATHS
 - (b) 16 answers of PHY
 - (c) 16 answers of CHY

Your task is to evaluate the response of each candidate

1. Compute the total marks obtained
2. Compute the marks obtained in MATHS
3. Compute the marks obtained in PHY
4. Compute the marks obtained in CHY
5. Store the above 4 in a `candidate_scores` table
6. You are advised to design the format of the `candidate_scores` table.

Perform the following tasks

Task 01 (10 marks) Create database and tables. Using C++ program and MySQL C API, perform the following:

1. Create a new database named `week12`
2. Switch to this database.

3. Create a table `solutions` with suitable columns to store: question number, subject, question type, correct answer, maximum marks
4. Create a table `candidate_responses` with suitable columns to store: registration number, followed by
 - 16 responses for MATHS
 - 16 responses for PHY
 - 16 responses for CHY
5. Create a table `candidate_scores` with suitable columns to store: registration number, maths marks, phy marks, chy marks, total marks.
6. Choose suitable primary keys and data types for each of the above table.

Task 02 (12 marks) Load the answer key. Using C++ program and MySQL C API:

1. Read the file `key.csv`
2. Insert all rows into the `solutions` table.
3. Ensure that all 48 solutions are loaded: (Perform this task using C++ MySQL APIs and by firing relevant SQL query)
 - 16 MATH keys
 - 16 PHY keys
 - 16 CHY keys

Task 03 (18 marks) Load the candidate responses. Using C++ program and MySQL C API:

1. Read the file `jee_candidates_1000.csv`
2. Insert these responses into the `candidate_responses` table.
3. Ensure (using C++ MySQL APIs and by firing relevant query) that each candidate should have exactly 48 responses.
4. Ensure (using C++ MySQL APIs and by firing relevant query) that there are a total of 1000 candidates responses.

Task 04 (40 marks) Evaluate every candidate's answers

1. Using C++ and MySQL C API, compare each candidate response with the official key and compute marks.
2. **Award maximum marks if the candidate response match the key. 0 otherwise.**
3. For each candidate:
 - (a) Evaluate all 16 MATHS answers and compute `maths_marks`
 - (b) Evaluate all 16 PHY answers and compute `phy_marks`.
 - (c) Evaluate all 16 CHY answers and compute `chy_marks`.
 - (d) Compute `total_marks = maths_marks + phy_marks + chy_marks`
4. Insert the computed answer into `candidate_scores` table.

Task 05 (10 marks) List marks of all students. Using C++ program and MySQL C API:

1. Retrieve all rows from `candidate_scores`.
2. Display the following for each candidate

- Registration number
- `maths_marks`
- `phy_marks`
- `chy_marks`
- `total_marks`

Example output: GHY1001,10,10,15,35

3. Print the records in descending order of `total_marks`. Hint: Perform sorting task as an SQL query within your C++ program

Proper Use (10 marks) for proper use MySQL C API result-processing functions.

Instructions Adhere to the following

1. Write the above in a `.cpp` file
2. You should upload the above `.cpp` file through MS assignments.
3. Mobile phones are not allowed inside the lab.